PROJECT REPORT

SSH BRUTE-FORCE ATTACK DETECTION USING SPLUNK

Project Domain: Cybersecurity and Security Information and Event Management (SIEM)

Project Title: SSH Brute-Force Attack Detection and Log Monitoring Using Splunk

Operating Systems: Kali Linux and Parrot OS

Tools Used: Splunk Enterprise, Splunk Universal Forwarder, OpenSSH, Linux Authentication Logs

Project Type: Cybersecurity Monitoring and Threat Detection

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1. ABSTRACT

SSH (Secure Shell) is a widely used protocol for remotely accessing and managing Linux systems. Attackers may attempt to gain unauthorized access by repeatedly trying different usernames and passwords. This technique is known as an SSH brute-force attack.

The main objective of this project is to detect suspicious SSH login activity using Splunk, a Security Information and Event Management (SIEM) platform. Kali Linux is used as the target system and Splunk monitoring server, while Parrot OS is used to simulate SSH login attempts in an isolated lab environment.

Linux authentication logs are collected from `/var/log/auth.log` and monitored using Splunk. Splunk Universal Forwarder is configured to collect and forward log events to Splunk Enterprise through TCP port 9997. Search Processing Language (SPL) queries are used to identify failed login attempts, analyze source IP addresses, count authentication failures, and detect repeated failures within a five-minute time window.

The project demonstrates how centralized log monitoring and simple detection rules can help security analysts identify suspicious authentication activity and improve security monitoring.

2. INTRODUCTION

Cybersecurity is the practice of protecting computer systems, networks, applications, and data from unauthorized access and malicious activities.

One common security threat is a brute-force attack. In this attack, an attacker repeatedly attempts to authenticate using different passwords or usernames until access is obtained or the attempts are blocked.

SSH is commonly used by system administrators to access Linux servers remotely. If SSH services are exposed without suitable security controls, repeated login attempts may increase the risk of unauthorized access.

Security teams use SIEM tools to collect, analyze, search, and monitor security events from different systems. Splunk Enterprise is a SIEM and data analytics platform that can search machine-generated data and help identify suspicious patterns.

This project implements a basic SSH authentication monitoring solution using Splunk and Linux authentication logs.

3. PROBLEM STATEMENT

Repeated SSH authentication failures can indicate password guessing, username enumeration, or other suspicious login activity. Manually reviewing authentication logs is time-consuming and may make it difficult to identify repeated attempts quickly.

Therefore, a log monitoring solution is required to collect SSH authentication events, search for failed logins, identify their source IP addresses, and flag repeated failures for further investigation.

This project addresses the problem by implementing an SSH log monitoring and detection workflow using Splunk Enterprise and Splunk Universal Forwarder.

4. OBJECTIVES

The primary objectives of this project are:

1. To understand SSH authentication and brute-force attack techniques.
2. To configure SSH connectivity between Kali Linux and Parrot OS in an isolated lab.
3. To identify failed SSH login attempts from Linux authentication logs.
4. To configure Splunk Enterprise for log searching and analysis.
5. To configure Splunk Universal Forwarder to monitor and forward authentication logs.

6. To use SPL queries to identify failed logins and suspicious source IP addresses.
7. To detect repeated authentication failures within a defined time window.
8. To understand how SIEM tools support security monitoring and incident investigation.

5. SCOPE OF THE PROJECT

The scope of this project includes:

- SSH authentication log collection.
- Monitoring failed login attempts.
- Identifying source IP addresses associated with failures.
- Counting failed authentication events.
- Detecting repeated failures within five-minute intervals.
- Searching and analyzing events using Splunk.
- Demonstrating basic security event investigation.

The project focuses on a controlled laboratory environment. It does not implement a complete enterprise security monitoring system or automatically block malicious IP addresses.

6. TOOLS AND TECHNOLOGIES

6.1 Kali Linux

Kali Linux is used as the target Linux system and Splunk monitoring host. The SSH service generates authentication logs when login attempts occur.

6.2 Parrot OS

Parrot OS is used as the test client to initiate controlled SSH login attempts against the Kali Linux system.

6.3 Splunk Enterprise

Splunk Enterprise is used to search, analyze, and visualize authentication logs. It allows security analysts to create SPL queries for identifying failed logins and suspicious patterns.

6.4 Splunk Universal Forwarder

Splunk Universal Forwarder is a lightweight agent that monitors configured log files and forwards events to a Splunk receiving instance.

In this lab, the Universal Forwarder is installed at `/opt/splunkforwarder`.

6.5 OpenSSH

OpenSSH provides the SSH server and client functionality used for remote access and authentication testing.

6.6 Linux Authentication Logs

The file `/var/log/auth.log` contains authentication-related events, including SSH login failures, successful authentications, and other security-related messages.

6.7 Splunk Search Processing Language (SPL)

SPL is used to search and analyze log events, extract source IP addresses, calculate event counts, and identify repeated authentication failures.

7. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or laptop capable of running virtual machines.
- Sufficient RAM and storage for both Linux environments and Splunk.
- Network connectivity between the virtual machines.

Software Requirements

- Kali Linux.
- Parrot OS.
- Splunk Enterprise.

- Splunk Universal Forwarder.
- OpenSSH server and client.
- Web browser for accessing Splunk Web.

Network Configuration

Component	Configuration
Target and Splunk host	Kali Linux
Kali IP address	`192.168.56.3`
Test client	Parrot OS
Parrot IP address	`192.168.56.4`
SSH port	TCP 22
Splunk receiving port	TCP 9997
Splunk Web port	TCP 8000
Authentication log	`/var/log/auth.log`

These IP addresses represent the lab configuration used in this project.

8. SYSTEM ARCHITECTURE

The project follows this workflow:

Parrot OS → SSH Login Attempts → Kali Linux SSH Server → Authentication Logs → Splunk Universal Forwarder → Splunk Enterprise → SPL Queries → Detection Results

Architecture Explanation

Step 1: Generate test events

Parrot OS initiates controlled SSH login attempts against the Kali Linux SSH server.

Step 2: Generate authentication logs

The Kali Linux SSH service records authentication failures in `/var/log/auth.log`.

Step 3: Collect log data

Splunk Universal Forwarder monitors the configured authentication log file.

Step 4: Forward events

The Universal Forwarder sends events to the Splunk receiving service through TCP port 9997.

Step 5: Index events

Splunk Enterprise receives and indexes the events in the configured index.

Step 6: Analyze events

SPL queries identify failed authentication attempts, extract source IP addresses, and count repeated failures.

Step 7: Detect suspicious activity

A detection query flags source IP addresses with three or more failed attempts within a five-minute interval for investigation.

9. IMPLEMENTATION METHODOLOGY

9.1 Configure the SSH Service

The SSH service was checked on Kali Linux to confirm that it was running and listening on TCP port 22.

Command:

```
sudo systemctl status ssh
```

To check the listening port:

```
sudo ss -lntp | grep ':22'
```

The SSH service must be active before authentication testing can take place.

9.2 Verify Network Connectivity

The network connection between Parrot OS and Kali Linux was tested.

From Parrot OS:

```
ping -c 4 192.168.56.3
```

The SSH port was checked using:

```
nc -vz -w 3 192.168.56.3 22
```

These checks confirmed connectivity between the lab systems.

9.3 Generate Controlled Failed Login Events

A test login attempt was initiated from Parrot OS using a deliberately invalid username.

```
ssh invaliduser@192.168.56.3
```

A few incorrect authentication attempts were performed in the authorized lab environment.

The test generated authentication failure events in the Kali Linux authentication log.

9.4 Verify Authentication Logs

The following command was used on Kali Linux to monitor new authentication events:

```
sudo tail -f /var/log/auth.log
```

To search specifically for authentication failures:

```
sudo grep -Ei \
'Failed password|Invalid user|authentication failure' \
/var/log/auth.log
```

The following event was observed:

```
2026-10-11T12:14:33.125023+05:30 kali sshd-session[54316]: Failed
password for invalid user invaliduser from 192.168.56.4 port 59860 ssh2
```

This event indicates that an SSH authentication attempt using the invalid username `invaliduser` failed. The source IP address was `192.168.56.4`, corresponding to Parrot OS in the lab.

Several failed authentication events were observed from the same source during the controlled test.

9.5 Configure Splunk Enterprise

Splunk Enterprise was installed under:

```
/opt/splunk
```

Splunk Web was accessed through:

```
http://127.0.0.1:8000
```

The receiving port was configured to accept forwarded events on TCP port 9997.

The receiving configuration can be checked in Splunk Web under:

Settings → Forwarding and receiving → Configure receiving.

9.6 Configure Splunk Universal Forwarder

The Universal Forwarder was installed under:

```
/opt/splunkforwarder
```

Its running status was checked using:

```
sudo /opt/splunkforwarder/bin/splunk status
```

The output confirmed that the Splunk daemon and helper processes were running.

The Forwarder destination was configured as follows:

```
sudo /opt/splunkforwarder/bin/splunk add forward-server 127.0.0.1:9997
```

The authentication log was configured for monitoring:

```
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -
index main -sourcetype auth
```

The configured destinations were checked using:

```
sudo /opt/splunkforwarder/bin/splunk list forward-server
```

The monitored files were checked using:

```
sudo /opt/splunkforwarder/bin/splunk list monitor
```

The monitoring configuration included `/var/log/auth.log`.

9.7 Search and Analyze Events in Splunk

Splunk Search & Reporting was used to execute SPL queries. The search time range was selected according to the test period.

The following sections describe the queries used for log analysis and suspicious activity detection.

10. SPL QUERIES USED IN THE PROJECT

10.1 Query 1: Verify Indexed Authentication Events

Purpose: To confirm that authentication events are available in Splunk and identify the index, host, source, and sourcetype.

```
index=* source="/var/log/auth.log"
| stats count by index host source sourcetype
```

Expected outcome: A summary of the available authentication events and their index configuration.

10.2 Query 2: Find Failed SSH Login Attempts

Purpose: To identify events containing failed SSH authentication messages.

```
index=* source="/var/log/auth.log" "Failed password"
| table _time host _raw
| sort - _time
```

Expected outcome: Events containing the `Failed password` message, including timestamps and source details.

10.3 Query 3: Investigate a Specific Source IP

Purpose: To investigate events associated with the Parrot OS IP address.

```
index=* source="/var/log/auth.log" "192.168.56.4"
| table _time host _raw
| sort - _time
```

Expected outcome: Authentication events and other relevant log entries containing the specified IP address.

10.4 Query 4: Count Failed Attempts by Source IP

Purpose: To identify IP addresses associated with the highest number of failed SSH authentication attempts.

```
index=* source="/var/log/auth.log" "Failed password"
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count AS failed_attempts by src_ip
| sort - failed_attempts
```

Explanation:

- `rex` extracts the source IP address.
- `stats count` calculates the number of matching events per IP.
- `sort` displays IP addresses with higher counts first.

Expected outcome: A table showing each source IP and its failed authentication count.

10.5 Query 5: Detect Repeated Failures Within Five Minutes

Purpose: To flag source IP addresses that generate three or more failed authentication events in the same five-minute time bucket.

```
index=* source="/var/log/auth.log" "Failed password"
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| bin _time span=5m
| stats count AS failed_attempts by _time src_ip
| where failed_attempts >= 3
| sort - failed_attempts
```

Explanation:

- `rex` extracts the source IP.
- `bin _time span=5m` groups events into five-minute buckets.
- `stats` calculates the failure count for each IP and time bucket.
- `where failed_attempts >= 3` filters the results to the configured threshold.

Expected outcome: Source IP addresses with at least three matching failed login events in a five-minute bucket.

This rule identifies suspicious activity for investigation. It does not, by itself, prove that an attack is malicious.

10.6 Query 6: Compare Failed and Successful SSH Authentication

Purpose: To distinguish failed password attempts from successful password authentication events.

```
index=* source="/var/log/auth.log"
("Failed password" OR "Accepted password")
| rex "(?:Failed|Accepted) password for (?:invalid user )?(?
<username>\S+) from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| eval login_status=if(
    like(_raw,"%Failed password%"),
    "Failed",
    "Successful"
  )
| stats count by src_ip username login_status
| sort - count
```

Expected outcome: A summary of matching failed and successful password authentication events, grouped by source IP, username, and status.

Note: This query covers password authentication messages. Other SSH authentication methods may require different searches.

11. RESULTS AND OBSERVATIONS

During the controlled lab test, the following observations were recorded.

Test	Observation
SSH service check	SSH service was listening on TCP port 22.
Network connectivity	Parrot OS could reach Kali Linux.
SSH authentication test	Invalid username login attempts generated failed authentication events.
Authentication log monitoring	Relevant events appeared in `/var/log/auth.log`.
Universal Forwarder status	The Forwarder daemon was running.
File monitoring	`/var/log/auth.log` appeared in the Forwarder's monitored files.
Splunk event search	SPL searches were prepared to locate and analyze authentication events.
Repeated failure detection	A threshold-based query was configured to identify three or more failures within a five-minute bucket.

The recorded authentication events included multiple failures for `invaliduser` from `192.168.56.4`. This demonstrates that the controlled SSH authentication test generated events suitable for security monitoring.

The final event ingestion status should be supported by Splunk screenshots showing indexed events and the results of the detection queries.

12. SECURITY ANALYSIS

The collected events can help a security analyst investigate suspicious SSH authentication activity.

The key indicators include:

1. **Repeated failed logins:** Multiple failures may indicate password guessing or incorrect credentials.

2. **Invalid usernames:** Attempts to authenticate with nonexistent usernames may indicate username enumeration or automated scanning.
3. **Source IP concentration:** A high number of failures from one IP can indicate concentrated login activity.
4. **Time-based patterns:** Repeated failures in a short interval can justify further investigation.
5. **Successful authentication after failures:** A successful login following several failures may warrant additional investigation, particularly if the account or source IP is unexpected.

A failed login alone does not establish malicious intent. Administrators must consider legitimate user errors, service configuration, source identity, account activity, and the wider context.

13. LIMITATIONS

The project has the following limitations:

- The experiment was conducted in a small lab environment rather than a production network.
- The detection rule uses a fixed threshold of three failed events in a five-minute bucket.
- The rule may flag legitimate users who repeatedly enter incorrect passwords.
- Distributed attempts from multiple IP addresses may require additional detection logic.
- Authentication events are useful only if logs are collected and indexed correctly.
- Missing permissions, receiver configuration problems, incorrect index settings, or time-range selection can prevent expected events from appearing in searches.
- The basic detection query does not automatically block source IP addresses or disable accounts.
- Running Splunk Enterprise and Universal Forwarder on the same host can result in duplicate ingestion if both independently monitor the same file.

14. FUTURE ENHANCEMENTS

The project can be improved through the following enhancements:

1. **Automated alerts:** Configure Splunk alerts when a source IP exceeds the defined failure threshold.
2. **Dashboards:** Create visualizations for failed logins, source IP activity, and authentication trends.
3. **Advanced correlation:** Correlate repeated failures with successful authentication events.
4. **Multiple log sources:** Monitor authentication events from additional Linux systems.
5. **Threat intelligence:** Enrich source IP addresses with trusted threat intelligence data where appropriate.
6. **Detection tuning:** Adjust thresholds based on normal system behavior to reduce false positives.

7. **Notification integration:** Send alerts to authorized security personnel through approved notification channels.
8. **Security controls:** Implement SSH key authentication, disable unnecessary remote access, and apply appropriate rate-limiting or access restrictions.
9. **Automated response:** Evaluate carefully controlled response mechanisms with safeguards against blocking legitimate administrators.

15. CONCLUSION

This project demonstrated the use of Splunk for monitoring and analyzing SSH authentication activity in a controlled Linux environment.

Kali Linux was used as the target system, while Parrot OS generated test SSH login attempts. The authentication logs recorded failed login events, including invalid usernames and source IP addresses. Splunk Enterprise and Universal Forwarder were configured to support log collection and analysis.

SPL queries were developed to search failed authentication events, investigate a specific source IP, calculate failure counts, and identify repeated failures within five-minute time windows.

The project provides practical experience in Linux log analysis, SIEM configuration, log forwarding, SPL, and basic security detection. It also demonstrates how repeated authentication failures can be investigated as potential security incidents.

The implemented solution is a foundational monitoring system. In a production environment, it should be combined with appropriate authentication controls, tuned detection rules, alerting, and incident response procedures.

16. REFERENCES

1. Splunk Documentation — Universal Forwarder: https://docs.splunk.com/Documentation/Forwarder
2. Splunk Documentation — Forwarding Data: https://docs.splunk.com/Documentation/Splunk/latest/Forwarding
3. Splunk Documentation — Search Reference: https://docs.splunk.com/Documentation/Splunk/latest/SearchReference
4. OpenSSH Manual Pages: https://www.openssh.com/manual.html
5. MITRE ATT&CK — Brute Force (T1110): https://attack.mitre.org/techniques/T1110/
6. MITRE ATT&CK — Remote Services: SSH (T1021.004): https://attack.mitre.org/techniques/T1021/004/

17. APPENDIX: SCREENSHOTS TO INCLUDE

Attach your own screenshots from the completed lab under the following captions:

- **Figure 1:** Kali Linux SSH service status.
- **Figure 2:** Network connectivity between Parrot OS and Kali Linux.
- **Figure 3:** SSH failed login events in `/var/log/auth.log`.
- **Figure 4:** Splunk Enterprise receiving configuration for TCP port 9997.
- **Figure 5:** Splunk Universal Forwarder status.
- **Figure 6:** Universal Forwarder destination and monitored log file.
- **Figure 7:** Authentication events indexed in Splunk.
- **Figure 8:** SPL search results showing failed SSH login attempts.
- **Figure 9:** Failed login counts grouped by source IP.
- **Figure 10:** Five-minute repeated-failure detection query results.

Final note: Only label a screenshot as a successful result if it actually shows that result. In particular, include evidence of indexed events and the detection query output to demonstrate end-to-end operation.